EXPENSE VOUCHER MANAGEMENT SYSTEM

Full Stack Developer Internship Assignment - Project Overview & Submission

Evaluator: **Prachay Securities Private Limited** | Candidate: **Yash Ranpariya**

GitHub Repository:	https://github.com/Yash-Ranpariya/voucher-create
Demo Video Walkthrough:	https://github.com/Yash-Ranpariya/voucher-create#demo-video-walkthrough
Application Tech Stack:	React 18 + Vite + TypeScript | Node.js + Express + Prisma ORM (SQLite / PostgreSQL)
Credentials (Password: Password@123):	Employee: **emp1@abc.com**, **emp2@abc.com** | Director: **director@abc.com** | Accounts: **accounts@abc.com**

1. Project Overview & Objective

Company ABC historically managed employee expense vouchers manually through paper receipts and physical sign-offs. This project delivers an end-to-end, digital, role-based **Expense Voucher Approval & Reimbursement System** that automates voucher creation, dual-tier digital signatures, hierarchy approval rules, and financial ledger monitoring.

Three Core Stakeholder Roles:

- **Employee:** Create draft vouchers, upload receipt attachments with OCR scanning, draw/upload signature, submit for approval, track real-time reimbursement status.
- **Director (Admin):** Company-wide view of all submitted vouchers, filter by department/status, review expense receipts, approve with digital signature or reject with mandatory rejection remarks.
- **Accounts Team:** Organization-wide visibility, multi-field search & filtering (Bonus Point), verification of high-value vouchers, payment disbursement via Razorpay gateway, and CSV financial exports.

2. Special Features & Innovations (What Makes This Project Stand Out)

- **1. Dynamic Hierarchy Approval Rule (INR 50,000+ Gateway):** Vouchers below INR 50,000 go straight to Director for approval. Vouchers of INR 50,000 or greater automatically trigger *PENDING_ACCOUNTS_VERIFICATION* first so the finance team can audit budget allocations before Director sign-off.
- **2. Dual Digital Signatures (Canvas Pad + Image Upload):** Full signature compliance: Employees and Directors can either physically sign on an interactive canvas pad or upload an image file. Signatures are snapshotted permanently to the voucher audit record.
- **3. AI Receipt OCR Scanner (Tesseract.js):** Smart expense creation: Employees can upload a physical receipt photo, and built-in OCR automatically extracts the total amount, expense date, category, and expense title to save manual effort.
- **4. Payment Gateway Integration (Razorpay SDK):** Accounts department can trigger direct online reimbursement payouts through an integrated Razorpay payment modal with transaction IDs logged directly in the ledger.

• **5. State-of-the-Art Fintech Glassmorphism UI:** Custom-crafted dark aesthetic with multi-layer mouse parallax background (vintage ticket watermark motifs), interactive Recharts donut distributions, and smooth Framer Motion transitions.

• **6. Complete Audit Trail & Notifications:** Every action (Draft Created, Updated, Submitted, Approved, Rejected, Paid) is chronologically stamped in the *VoucherActivity* log alongside in-app alerts and test emails.

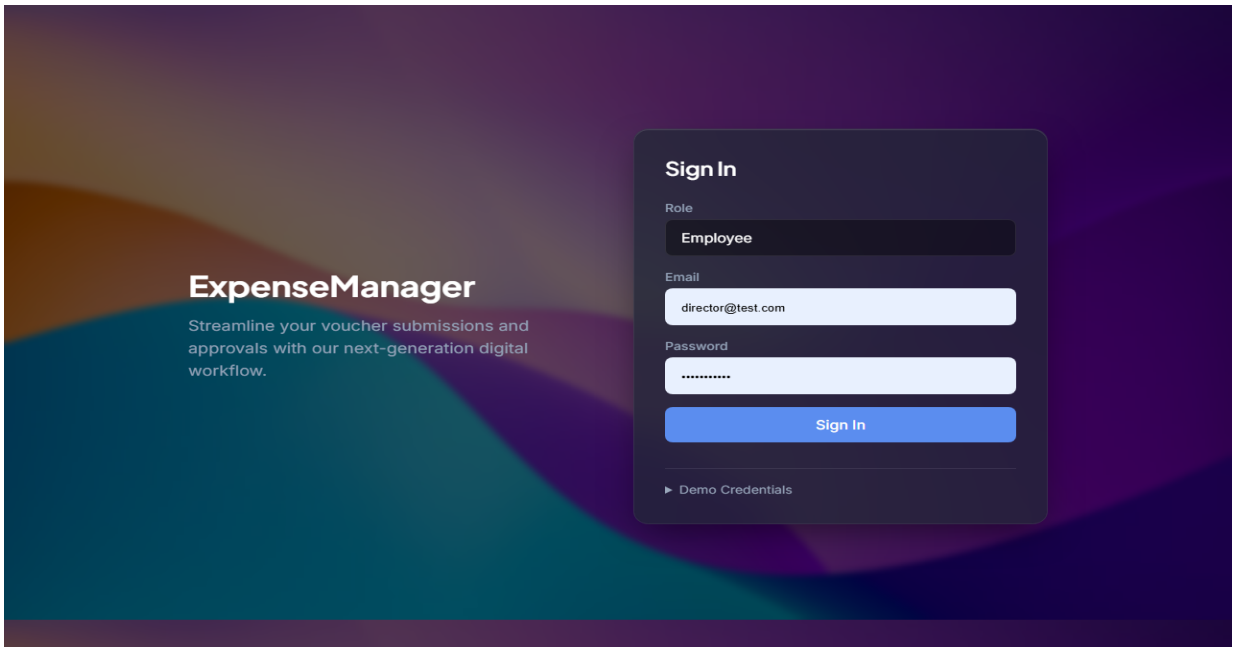
3. Step-by-Step Voucher Lifecycle

Stage	Trigger / Actor	Resulting Status & Logic
1. Create Draft	Employee	**DRAFT**. Fully editable and deletable by the employee.
2. Submit	Employee + Signature	If amount < INR 50k: **PENDING_APPROVAL**. If amount >= INR 50k: **PENDING_ACCOUNTS_VERIFICATION**.
3. Budget Check	Accounts Team	Verifies allocation: transitions to **PENDING_APPROVAL**.
4. Decision	Director + Signature / Reason	**APPROVED** (with signature) OR **REJECTED** (with mandatory reason).
5. Disbursement	Accounts Team	Processes payment via Razorpay: **PAID** (read-only final record).

4. Application Visual Showcase (Screenshots)

High-fidelity captures from the live running application demonstrating the glassmorphism design and role interfaces:

Figure 1: Role-Based Authentication & Sign In Screen



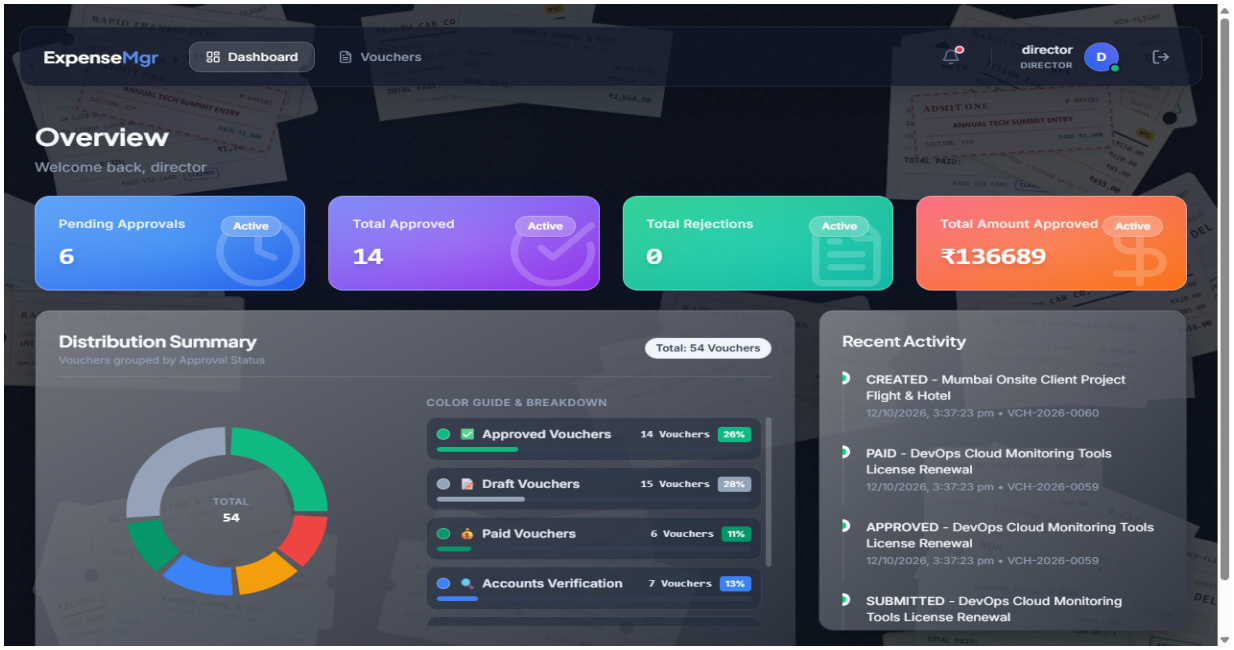
Glassmorphism sign-in card with role selector (Employee / Director / Accounts) and demo credentials helper.

Figure 2: Employee Dashboard (emp1 & emp2)



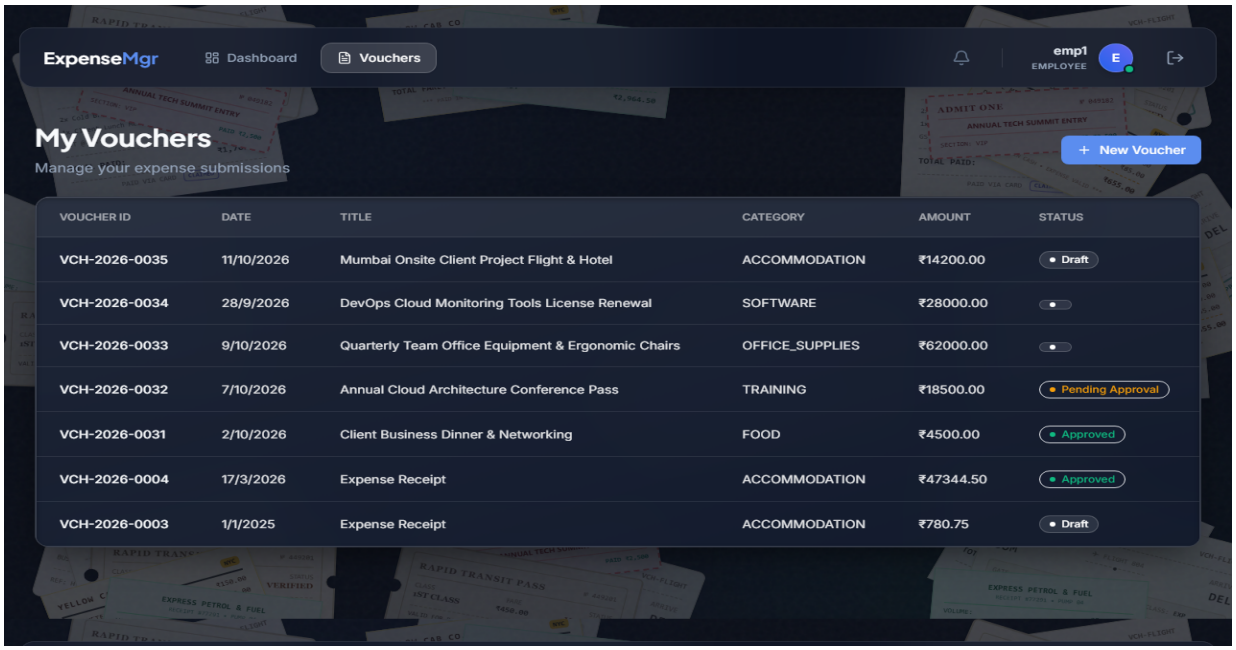
Real-time expense distribution donut chart with interactive color guide, metric cards, and personal voucher history.

Figure 3: Director Management & Approval Dashboard



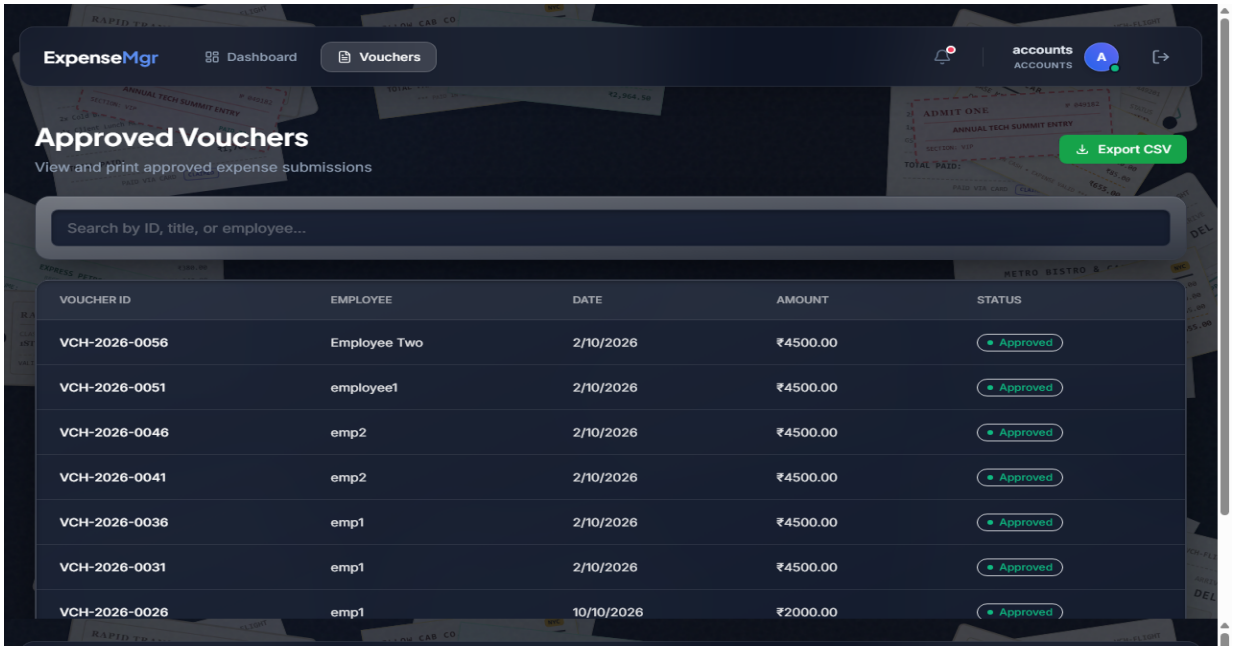
Company-wide overview: Pending approval count, approved amounts, category breakdowns, and audit feed.

Figure 4: Voucher Tracking Table (Draft, Pending, Approved, Paid)



Interactive voucher listing table showing real-time statuses, auto-generated IDs (VCH-2026-XXXX), and action buttons.

Figure 5: Accounts Team Monitoring & Export Hub



Finance team portal equipped with instant search, multi-column filters, Razorpay payment triggers, and CSV export.

5. Demo Video Walkthrough & Repository Submission

A complete video demonstration of the workflow has been organized into the GitHub repository documentation. You can review the full interactive walkthrough, test logins, signature signing, and voucher state transitions below:

Direct GitHub Repo:	https://github.com/Yash-Ranpariya/voucher-create
Demo Video Link:	https://github.com/Yash-Ranpariya/voucher-create#demo-video-walkthrough (Direct video stream & animated demo)
Local Setup:	Backend: npm run dev on port 5000 | Frontend: npm run dev on port 5173
Candidate Name:	**Yash Ranpariya** (Full Stack Developer Internship Candidate)

Evaluation Summary: The project strictly adheres to all requirements from the assignment specification (RBAC security, audit trails, responsive glassmorphism design, digital signatures, validation rules, and automated unique voucher number generation).